FEA STUDIO · VERSION 1.4

Testing Guide

Benchmark problems with known results: workflow in the program and expected values

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Edition November 11, 2026 · FEA Studio 1.4

Contents

01 Examples and tour 2

02 Testing (problems with known results) 13

CHAPTER 1

01 Examples and guide

Ten ready-made projects showing the program's capabilities. The guide opens them and explains step by step what is set and why.

Starting the guide

- **Example 1 – printed bracket**: a single PLA part; 3D printing and creep (14 steps).
- **Example 2 – steel + print assembly**: bolts, welds, heat, all kinds of supports and loads (12 steps).
- **Example 3 – flange joint**: pipe with cover, 4× M10 bolts, internal pressure 2.5 MPa, VDI 2230 (7 steps).
- **Example 4 – welded boom**: S355, 5 a4 fillet welds, 5000 N, weld fatigue per EN 1993-1-8 and IIW (7 steps).
- **Example 5 – electronics heat sink**: finned aluminium + stainless module, 40 W, convection, steady and transient heat, thermal stress (6 steps).
- **Example 6 – hall frame of beams**: beams without CAD, EN sections, snow and wind, natural frequencies, EN 1993-1-1 (5 steps).
- **Example 7 – dynamics of a PCB fixture**: point and surface mass, natural frequencies, harmonic 1 g, random vibration (ASD), probes (9 steps).
- **Example 8 – crank mechanism**: bodies from CAD, joints from geometry, drive, animation, reactions, transfer of the maximum instant to FEA (8 steps).
- **Example 9 – bolted column base (contact)**: frictional contact at the bolted interface, preload, joint opening, nonlinear sequence (8 steps).
- **Example 10 – bracket, topology optimization**: lightening with BESO, final shape and density in 3D, STL export (7 steps).

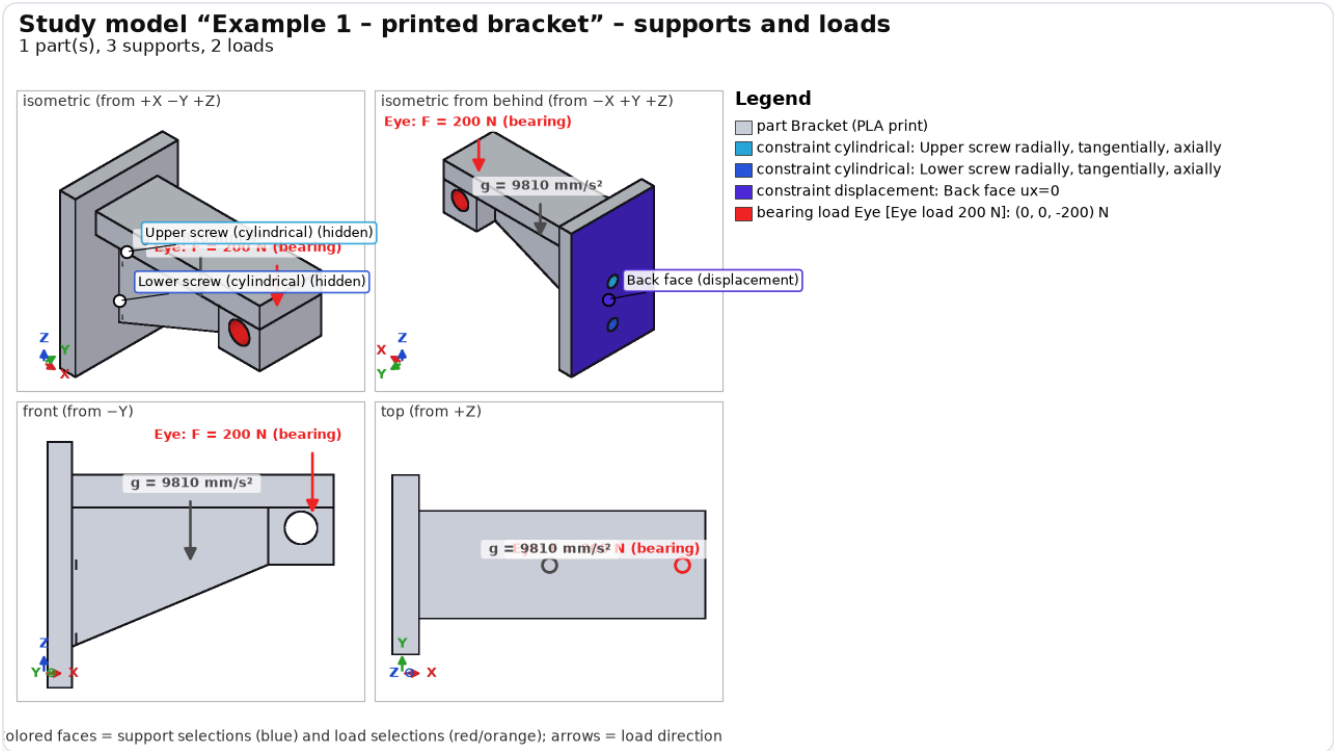
The guide is the panel on the right: **Show in app** switches the program to the state the step describes, **Next** / **Back** page through the steps, **Help for the step** opens the detailed page. You can work freely between steps.

> **Tip:** the guide opens a *copy* in `%LOCALAPPDATA%\FEAStudio\ukazky`; `Ctrl+S` does not overwrite the original.

Without the guide: `Ctrl+O` → `examples` in the installation → `example1_bracket.fea` to `example10_bracket_topology.fea`.

Example 1 – printed bracket

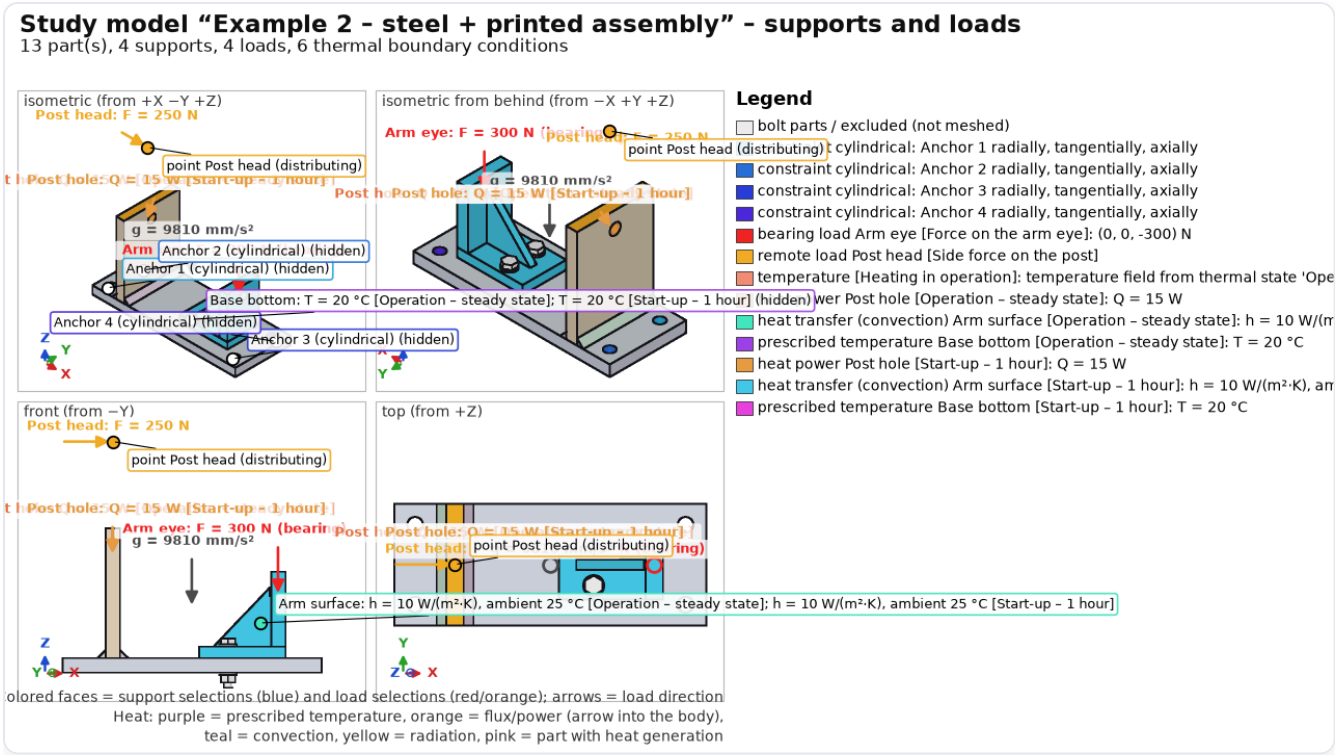
PLA bracket (Bambu Basic), build direction Z: a vertical plate 60×40×6 mm with two Ø5 screw holes, a horizontal arm 64×24×8 mm, and at its end a boss with an Ø8 eye (load pin) and a triangular rib.



Step	Specification
Materials	Print PLA Bambu Basic (E 2680 / E_z 2160 MPa, Rm 39 / Rm_z 35 MPa)
3D printing	build Z, layer 0.2 mm, line 0.42 mm, 3 perimeters, 5/3 solid layers, gyroid 20 %; Tsai-Wu; creep 1000 h at 30 °C for the Eye load case
Supports	cylindrical support of each screw hole (Upper screw, Lower screw), back face ux = 0
Loads	Eye load 200 N (bearing load, downward); Self-weight
Mesh	2.5 mm, tet10
Fatigue	sine 0–1 × Eye load (pulsating 0–200 N), 100 000 repetitions

Example 2 – steel + print assembly

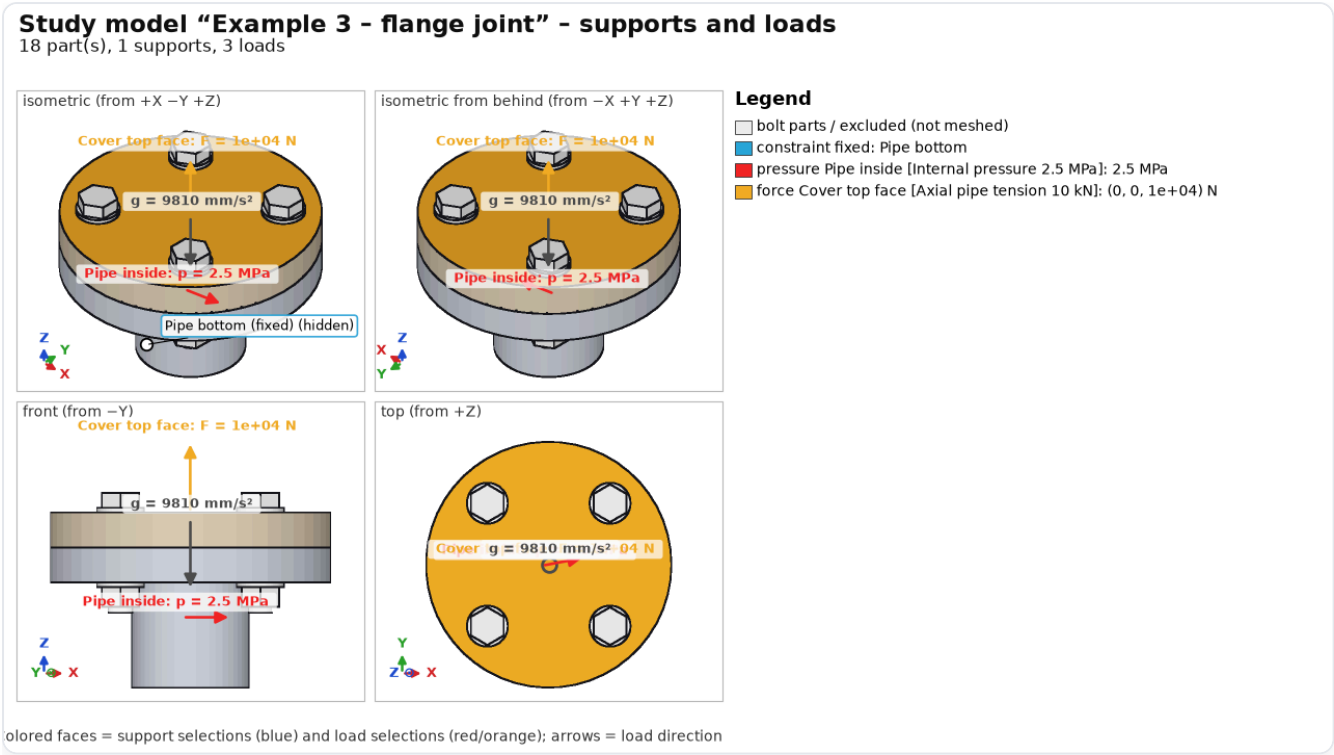
Steel base plate 180×70×10 mm (S235) with 4 anchor holes Ø9, column 10×70×90 mm welded by two a4 fillet welds, and a printed PLA arm bolted to the plate with 2× ISO 4017 M6 x 30 - 8.8, ISO 7089 washers and ISO 4032 nuts.



Step	Specification
Materials	Steel S235 (default), arm Print PLA Bambu Basic
3D printing	arm: 3 perimeters, gyroid 25 %, creep 2000 h / 30 °C for Force on the arm eye
Joints	2 M6 8.8 bolts, automatic preload (Bolt preload case); 2 a4 fillet welds (EN 1993-1-8, fatigue of root and toe)
Supports	cylindrical support of the 4 anchor holes
Heat	15 W into the column hole; arm convection 10 W/(m²·K) to 25 °C; base plate bottom 20 °C; steady state and 1 h transient ramp-up
Loads	Force on the arm eye 300 N (bearing); lateral force 250 N on the column at remote point [35; 35; 150]; Self-weight; Heating in operation (thermal stress from the steady state)
Mesh	4 mm; local refinement of the arm eye 1.5 mm
Fatigue	two sine channels shifted by 90° (non-proportional), 200 000 repetitions

Example 3 – flange joint

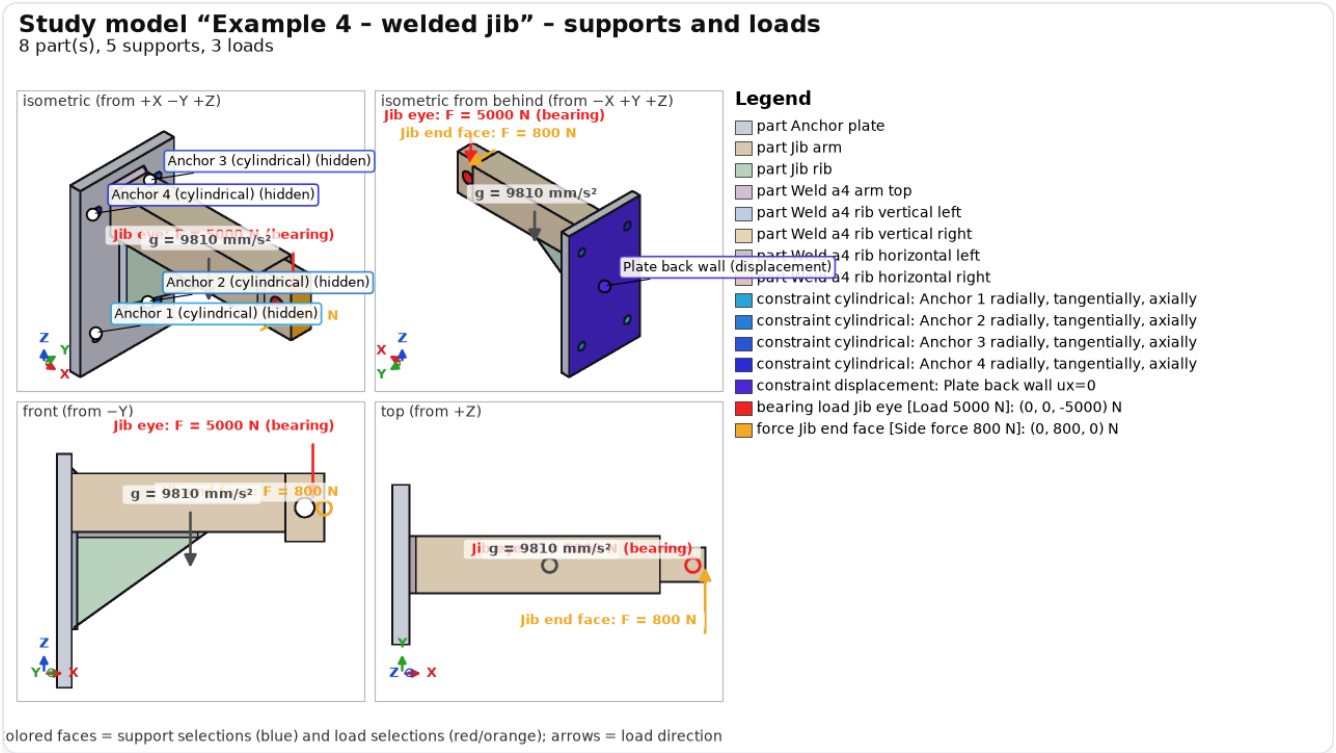
Pressure pipe Ø50/36 with a welded flange Ø120 and a solid cover, connected by 4× ISO 4017 M10 × 45 8.8 bolts with washers and nuts. It verifies the preload computation and the **VDI 2230** assessment.



Step	Specification
Materials	Steel S355 (default for flange and cover)
Joints	4 M10 8.8 bolts, automatic preload (tightening factor alpha_A = 1.6, friction μ = 0.12)
Supports	fixed support on the bottom end face of the pipe
Loads	internal pressure 2.5 MPa on the inner cylinder; axial tension 10 kN on the top end face of the cover; Self-weight; Bolt preload
Mesh	5.0 mm, tet10
Assessment	VDI 2230: assembly preload, tightening torque, safety against yielding and joint opening

Example 4 – welded boom

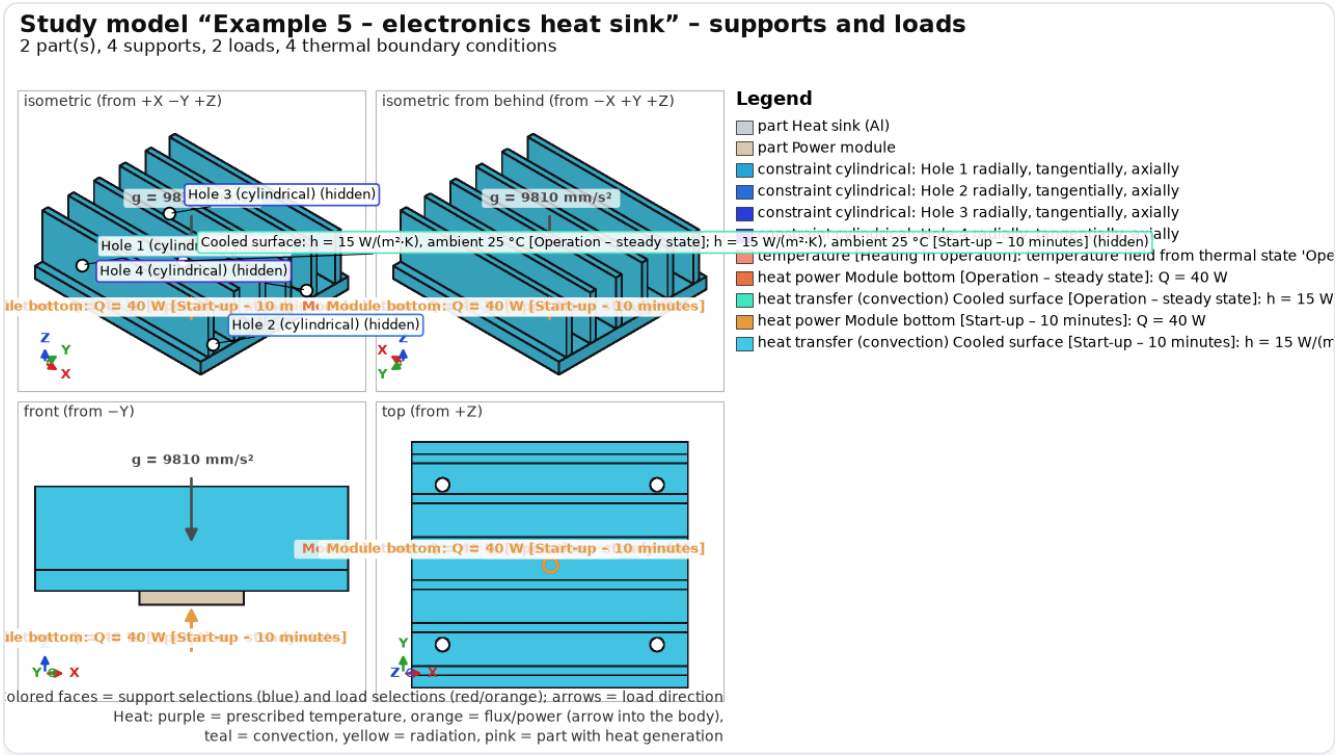
Welded cantilever boom: anchor plate with 4 holes Ø13, horizontal arm 50×60 × 220 mm with an eye for the 5000 N load pin, triangular rib 12 mm thick and 5 a4 fillet welds.



Step	Specification
Materials	Steel S355 (default)
Joints	5 a4 fillet welds (arm to plate at the top, rib welds to the plate and arm)
Supports	cylindrical support of the 4 anchor holes; back wall of the anchor plate ux = 0 (resting against a wall)
Loads	load 5000 N in the eye (bearing load); lateral force 800 N on the arm end face; Self-weight
Mesh	6.0 mm, tet10
Fatigue	cyclic lift 0–5 kN (200 000 cycles) shifted against the lateral force; IIW and EN 1993-1-8

Example 5 – electronics heat sink

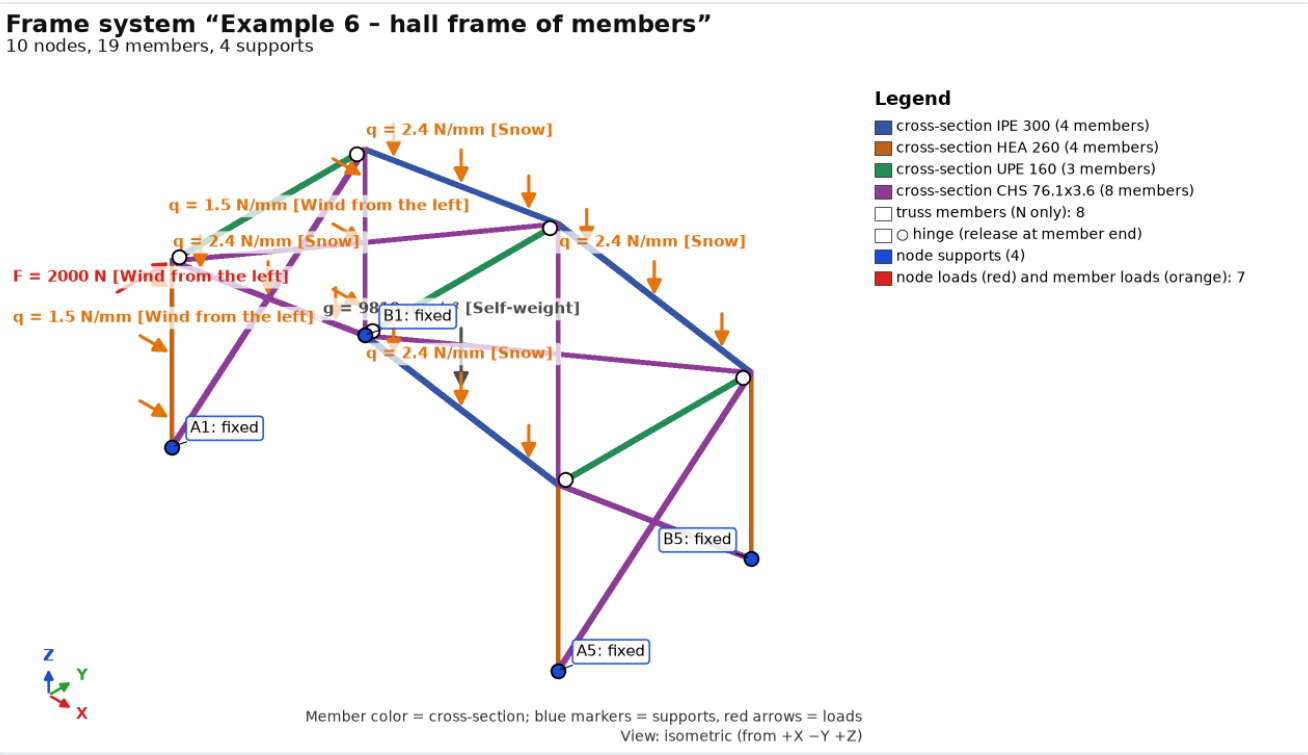
Aluminium finned heat sink 90×80×30 mm with 6 fins and a stainless power module 30×30×4 mm dissipating 40 W.



Step	Specification
Materials	Aluminium EN AW-6082 T6 (heat sink, k = 170 W/(m·K)); stainless steel 1.4301 (module, k = 15 W/(m·K))
Supports	cylindrical support of the 4 mounting holes Ø4.5 mm
Heat	40 W into the module; natural convection h = 15 W/(m²·K) on the fins to 25 °C; steady state and 10 min transient ramp-up
Mechanics	Heating in operation (temperature field from the steady state → bimetallic bending and stress); Self-weight
Mesh	4.0 mm, tet10

Example 6 – hall frame of beams

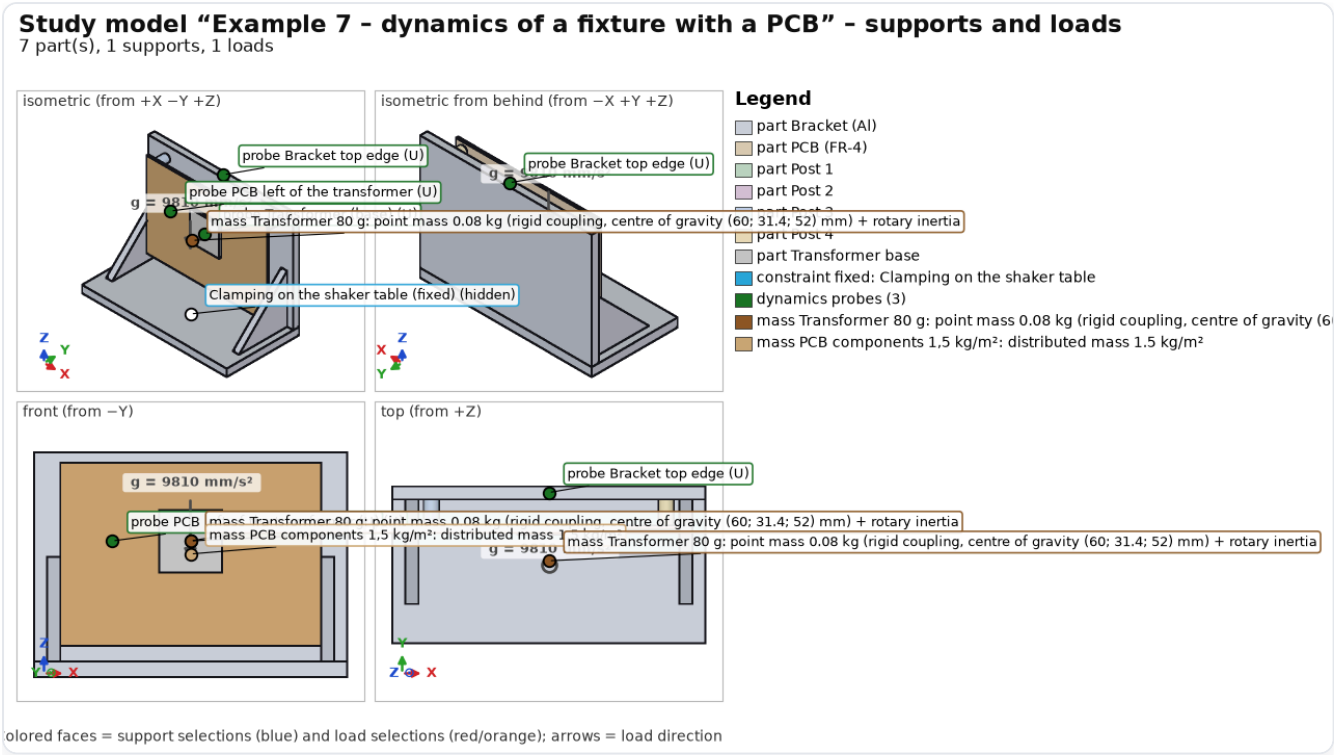
Steel hall made of beams only (study without STEP geometry): two frames, purlins and bracing. Results are not stored in the example; they are computed in seconds.



Step	Specification
Beams	columns HEA 260, rafters IPE 300 (span 12 m, gable roof), purlins UPE 160 pinned, bracing CHS 76.1×3.6 (axial only), fixed column bases; EN 1993-1-1 (column buckling lengths)
Loads	Self-weight; snow 0.8 kN/m² on rafters; wind from the left on columns + 2 kN in the gable
Solution	statics + 6 natural frequencies (1st ~4.8 Hz)

Example 7 – dynamics of a PCB fixture

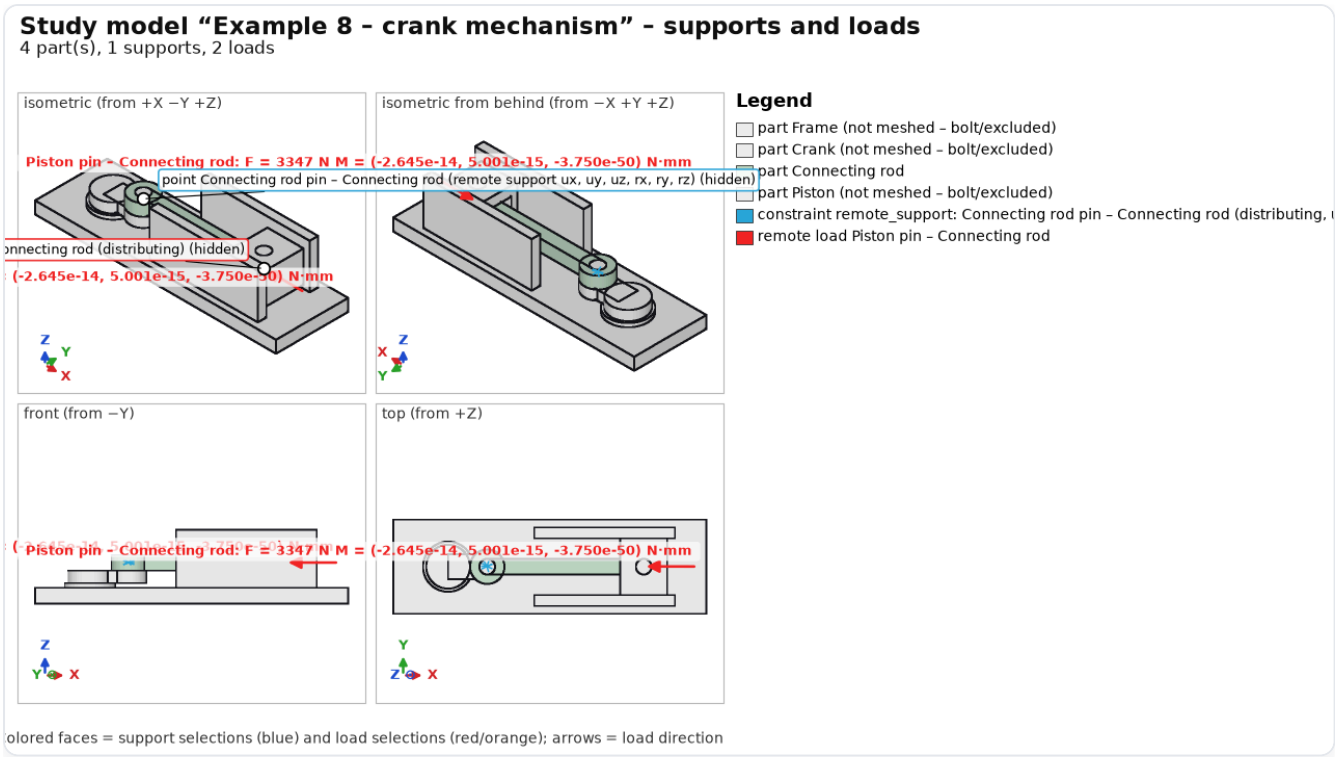
Aluminium bracket clamped to a shaker table, 100×70×1.6 mm FR-4 board on 4 brass standoffs Ø6, an 80 g transformer and the board population.



Step	Specification
Materials	Aluminium EN AW-6082 T6; brass CuZn39Pb3; custom “DPS FR-4 (orientačně)” (E 22 GPa, ρ 1,900 kg/m^3)
Supports	fixed bottom face of the base (= excited base)
Dynamics	80 g point mass with rotary inertia (rigid coupling to the footprint); 1.5 kg/m^2 surface mass on the board; 20 modes, ζ = 3 %; harmonic sine 1 g in Y 10–2,000 Hz; random ASD 20 Hz 0.008 → 50–500 Hz 0.02 → 2,000 Hz 0.005 g^2/Hz (4.8 g_{rms}, 0.5 h, Dirlik) – **an EXAMPLE in the structure of IEC 60068-2-64, not values from the standard**; probes on footprint, board and bracket
Mesh	6 mm, 3 mm at the board (local refinement), tet10, 32 k nodes; about 1 min

Example 8 – crank mechanism

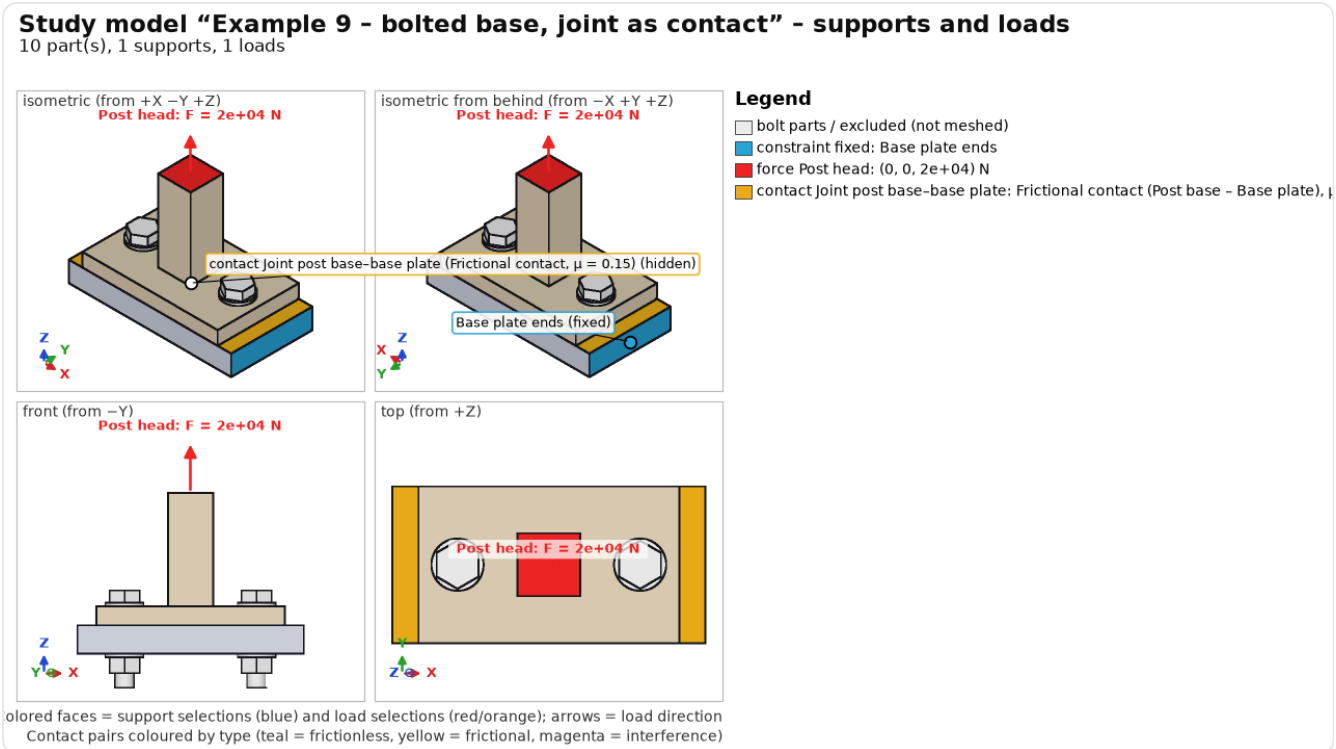
Frame with a bearing and a guide; crank r = 25 mm, connecting rod l = 100 mm, piston (slider); motor 3,000 rpm, gas force 3,000 N.



Step	Specification
Materials	Steel C45 +N (frame, crank, piston); connecting rod Steel 42CrMo4 +QT
Motion	4 bodies (fixed frame); 3 revolute joints proposed from geometry – the rod pins changed to ball + universal (non-redundant set); sliding guide of the piston; drive at constant 314.16 rad/s; 3,000 N on the piston; gravity; kinematics of 1 revolution, 121 outputs
Transfer to FEA	instant of maximum reaction in the piston pin → load case of the rod: remote load on eye B, inertia (Inertia), eye A on a distributing remote support (inertia relief); frame, crank and piston excluded
Mesh	2 mm, connecting rod only

Example 9 – bolted column base (contact)

Column base plate 100×60×10 with a 24×24 column on a 120×60×15 base plate, 2× ISO 4017 M10 × 45 8.8 with washers and nuts.

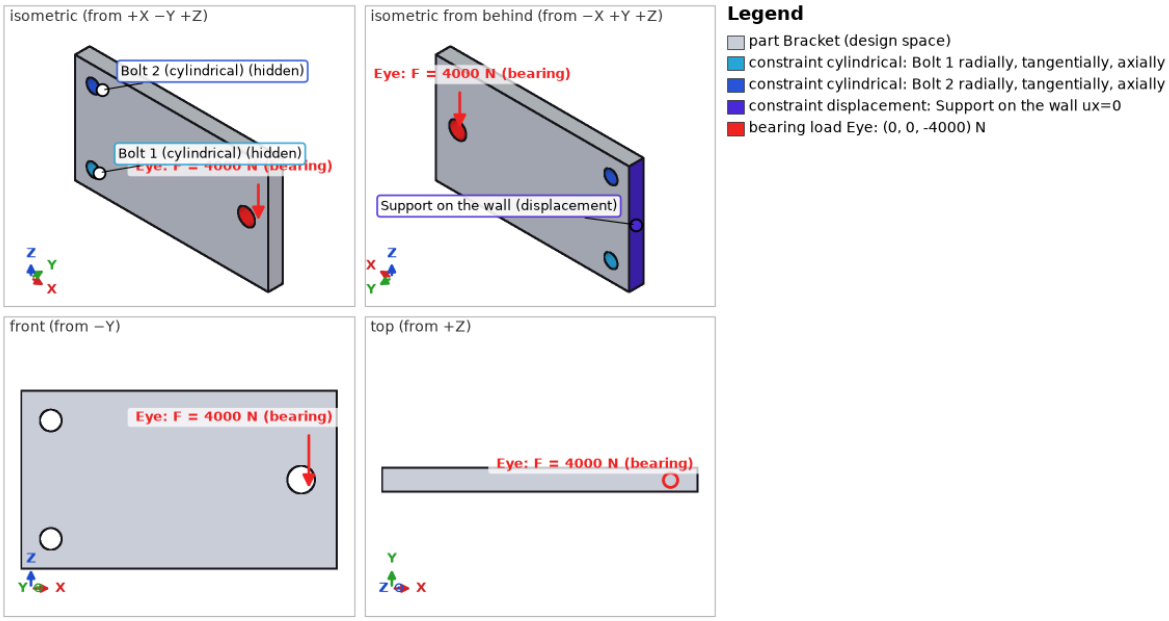


Step	Specification
Joints	2 bolts M10 8.8, automatic preload (VDI 2230)
Contacts	bolted interfaces as frictional contact μ = μ_T = 0.15 (penalty K = 10·E); nonlinear, as a sequence: bolt preload → tension 20 kN
Supports	base plate fixed at the ends x = ±60
Loads	tension 20 kN on the top of the column (z = 70)
Mesh	6 mm, tet10, conformal mesh with a split interface

Example 10 – bracket, topology optimization

Full steel plate 160×90×12 mm (S235) as the design space: two Ø11 holes for M10 bolts into a wall (x = 15 mm, z = 15 and 75 mm) and an Ø14 eye at the free end (x = 142 mm, z = 45 mm). Topology optimization removes lightly loaded material and keeps a shape that carries the load at the given volume.

Study model “Example 10 - bracket, topology optimization” - supports and loads
1 part(s), 3 supports, 1 loads



Colored faces = support selections (blue) and load selections (red/orange); arrows = load direction

Step	Setup
Materials	Steel S235 (isotropic elastic – a requirement of topology optimization)
Supports	cylindrical support of the holes Bolt 1 and Bolt 2; face Support at wall ux = 0
Loads	case Load 4 kN – bearing load in the eye, downward
Mesh	4 mm, tet10
Results	statics of the full design and the result (quantities Topology – density, Topology – final shape)
Topology (BESO)	target 40 % volume, filter R 8 mm, ER 4 %/step, at most 60 iterations, all selections protected (holes, eye, wall support); see Results – Topology optimization; result: 40 iterations, volume 40.0 %, compliance 2.2× the full design

Recreating the examples

A script creates the examples (geometry via OCCT in Python, the study via the program core):

```
.venv\Scripts\python.exe -m examples.create --solve
.venv\Scripts\python.exe -m examples.create --solve --only 8      (one example only)
.venv\Scripts\python.exe -m examples.create --solve --only 10     (example 10: topology, dozens of ccx
runs)
.venv\Scripts\python.exe -m examples.frames                      (example 6)
```

Without `--solve` only the input is saved. `examples\create.py` is also a template for building studies from Python – see Automation.

CHAPTER 2

02 Testing the program

Ten test problems with a known result (analytical solution or code-based calculation) and the example tours. Each has a procedure and a table of expected values.

Preparation

- Launch: the **FEA Studio** desktop shortcut or `FEA Studio.bat` in the program folder.
- Test models: `examples\user_tests` in the installation (`T1_…step` to `T10_…step`); recreate with `.venv\Scripts\python.exe -m examples.user_tests`.
- Guided examples: **Help** → **Example tour** (see Examples and tours).
- Verification of the whole program without the GUI (about 10 minutes): `.venv\Scripts\python.exe -m verification.run_all`. The output must end with **VÝSLEDEK: všechny kontroly prošly** (RESULT: all checks passed; the console is in Czech).

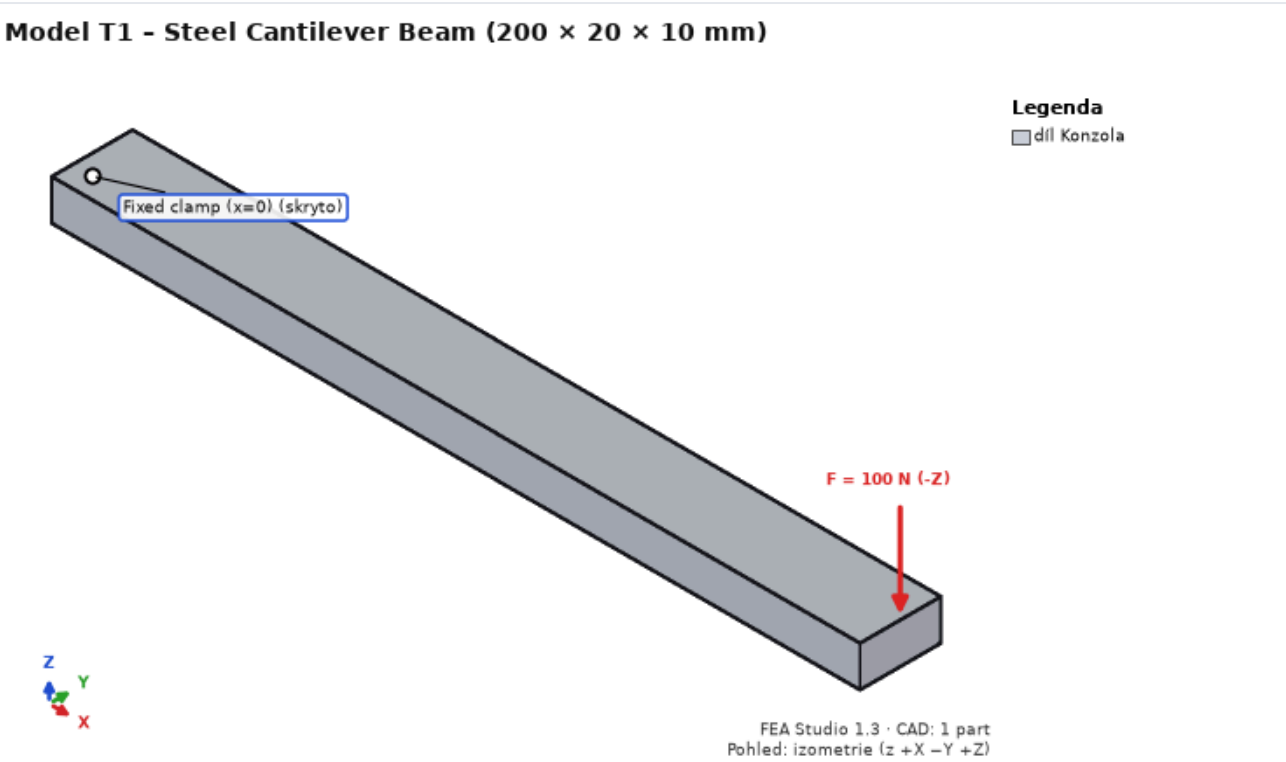
> **Tip:** FEA values differ from analytical ones by the mesh error; each value has a tolerance. Displacements within ~3 %, stresses away from singularities within ~5 %. Larger deviations are errors – send the .fea project and a description.

Recommended order

1. **Example tour 1** (printed bracket), about 1 min analysis.
2. **Example tour 2** (assembly: bolts, welds, heat, fatigue), about 3 min.
3. Tests T1–T7, 5–10 minutes each.
4. Dynamics, contacts and mechanisms: example tours 7–9 and tests T8–T10.
5. Your own STEP model – follow the step bar; `F1` = help.
6. Word report (`Ctrl+P`) and manual (Help → Example tour → Manual).

T1 – cantilever: statics, deflection and stress

Steel beam 200 × 20 × 10 mm, fixed at one end, 100 N force at the other, perpendicular to the 10 mm height.



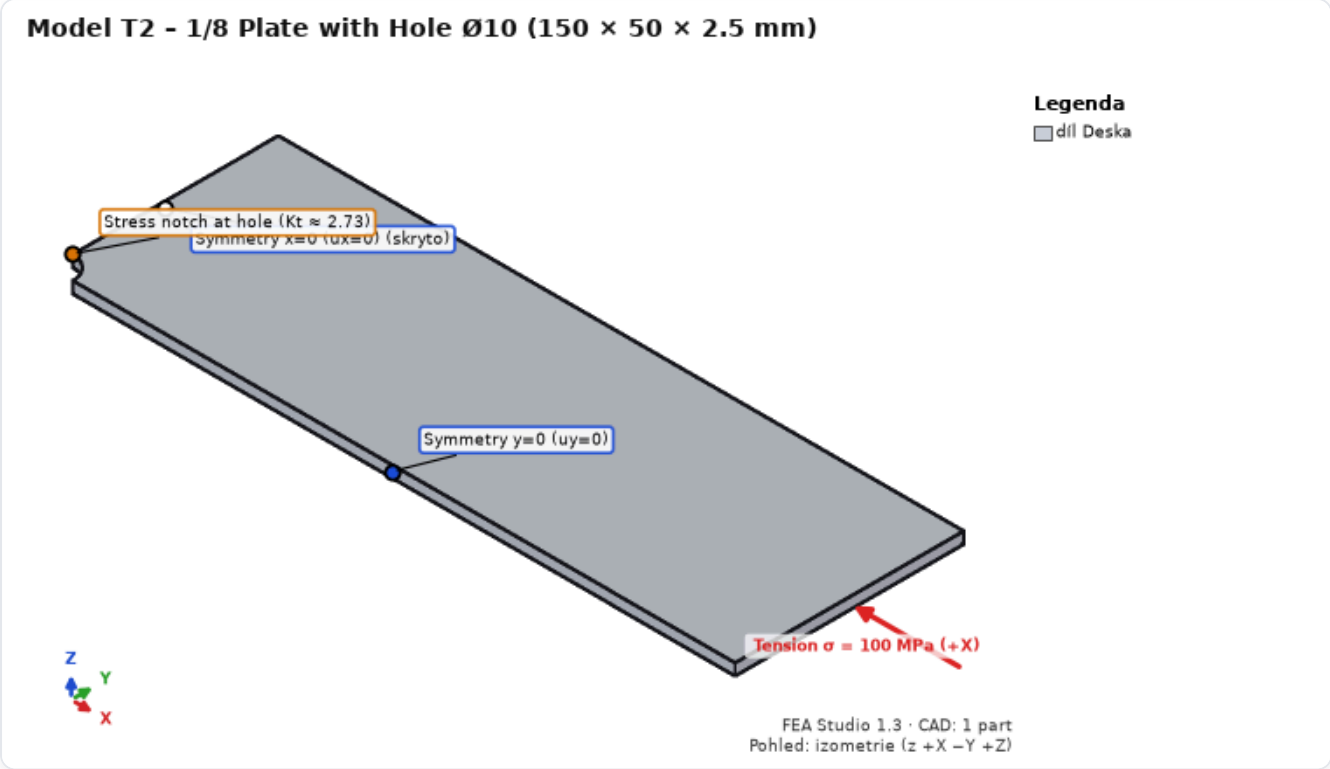
1. `Ctrl+O` → `T1_cantilever.step`.
2. Material of part Konzola: **Steel S235** (E = 210 000 MPa, ν = 0.3).
3. Supports: face x = 0 → **Fixed support**.
4. Loads: face x = 200 → **Force** (0; 0; −100) N.
5. Mesh: element size 2.5 mm → Generate mesh; Solve → Run analysis.

Quantity	Expected value	Where to read
Tip deflection (displacement Z)	0.763 mm (Timoshenko), FEA ~0.76 mm (±3 %)	Results → Displacement Z, summary
σx at mid-length, top / bottom face	+30 / −30 MPa (±5 %)	Stress σx, click at x = 100
Reaction Fz at the support	100 N	summary – reactions
Max. von Mises	at the support, marked as singularity (support edge)	Highest stresses

Fatigue on the same model: Fatigue step → channel of the force case: sine, mean 0, amplitude 1, 1000 cycles per block, 100 blocks, signed von Mises, steel S-N: SD = 20 MPa, ND = 10^6, k = 5, k2 = 0. At a node on the top face at x = 100: amplitude 30 MPa (±5 %), damage over 100 blocks ≈ 0.74, safety factor ≈ 1.06.

T2 – plate with a hole: stress concentration and symmetry

One eighth of a plate (width 100, thickness 5 mm, hole Ø10, tension 100 MPa): a 150 × 50 × 2.5 mm model with a quarter hole in the corner.

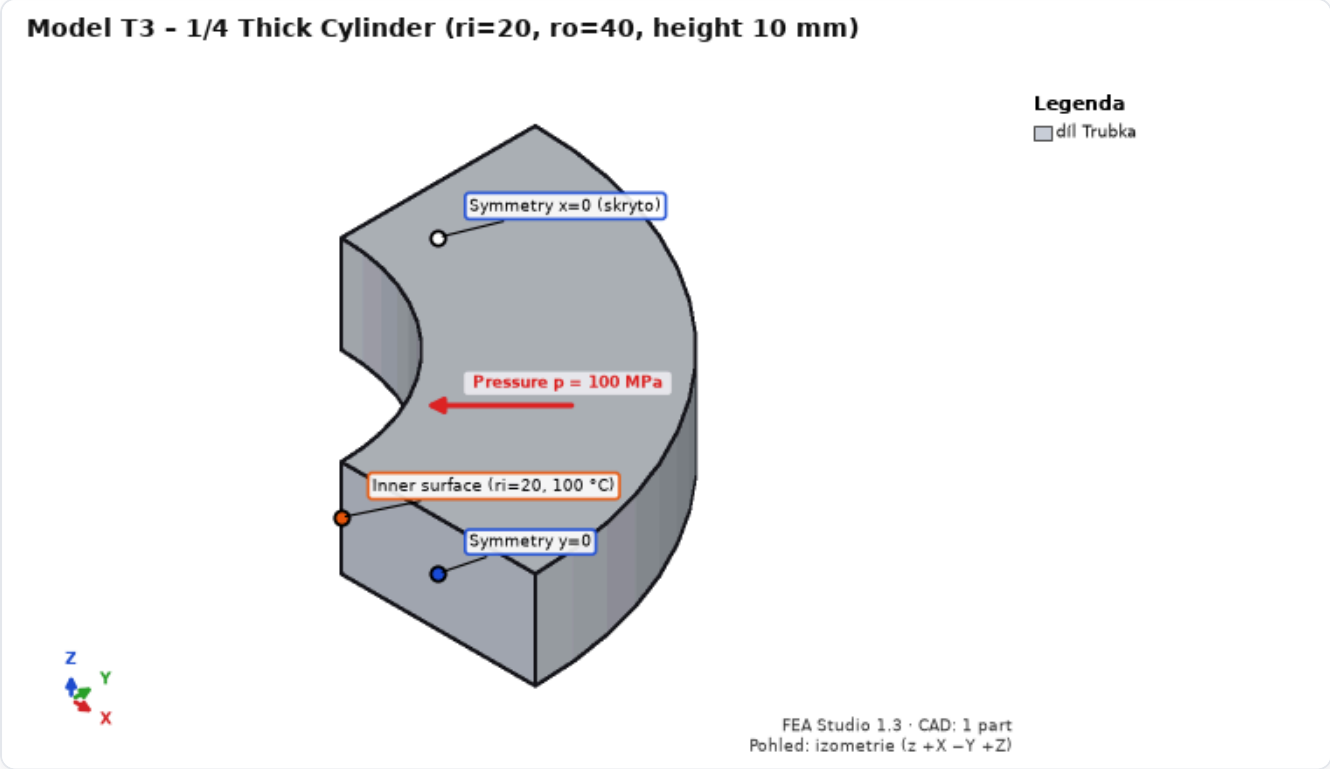


1. Open `T2_plate_with_hole_eighth.step`, Steel S235.
2. Symmetry supports (**Prescribed displacement** of one component): face x = 0 → ux = 0; face y = 0 → uy = 0; bottom face z = 0 → uz = 0.
3. Loads: end face x = 150 → **Pressure** −100 MPa (negative = tension).
4. Mesh 4 mm; local refinement of the hole face (Hole filter `H`) to 0.35 mm.

Quantity	Expected value	Where to read
σx max at the hole edge (x = 0, y = 5)	≈ 303 MPa (Kt,n = 2.729 × 111.1 MPa, ±5 %)	Stress σx
Reaction Fx in the plane x = 0	12 500 N (100 MPa × 50 × 2.5)	summary – reactions
Classification of the maximum	real location (not a singularity)	Highest stresses

T3 – thick-walled tube: Lamé, heat and thermal stress

Quarter of a ring, ri = 20, ro = 40 mm, height 10 mm.



1. Open `T3_pipe_quarter.step`, Steel S235.
2. Supports: face x = 0 → ux = 0; face y = 0 → uy = 0; bottom face z = 0 → uz = 0.
3. Loads: inner cylindrical face (R 20) → Pressure 100 MPa. Mesh 2.5 mm; run.

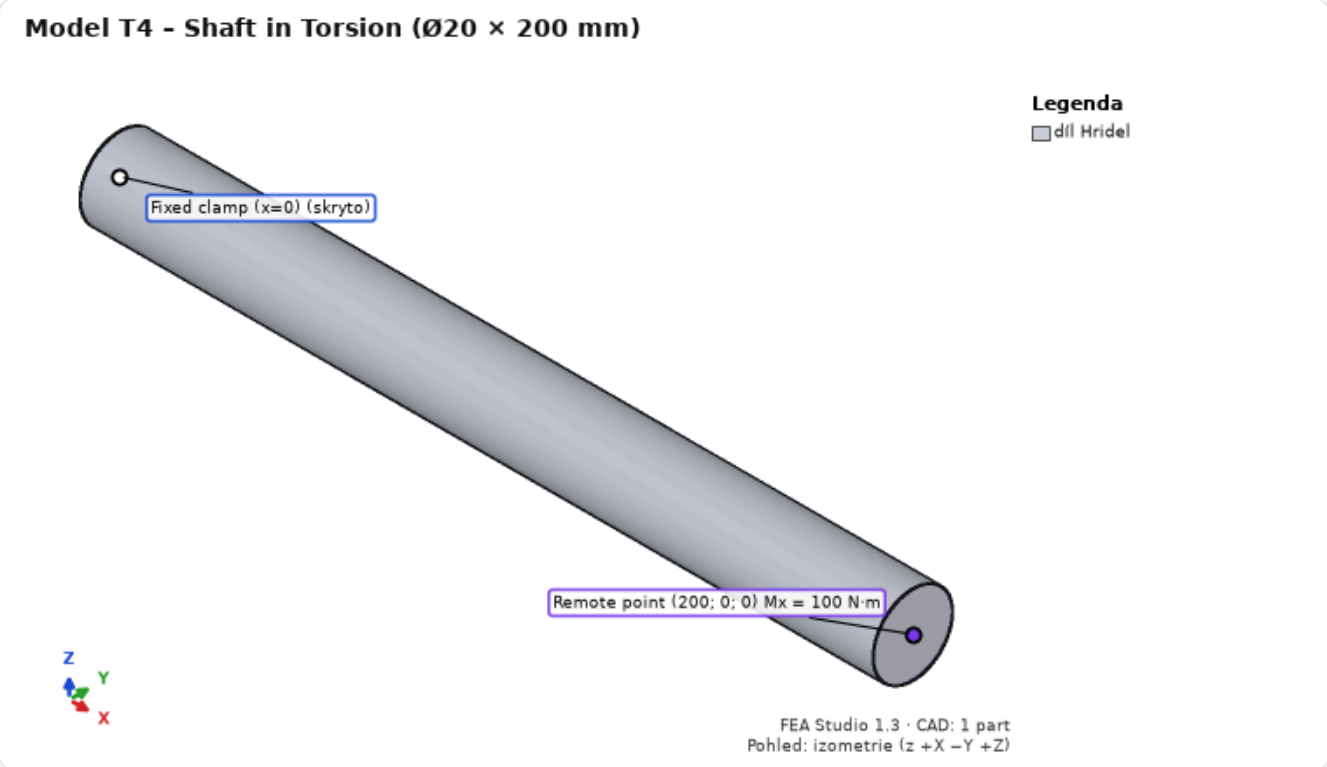
Quantity	Expected value
σθ (hoop) at ri / ro	166.7 / 66.7 MPa (±5 %)
von Mises at ri	233.3 MPa (±5 %)
Radial displacement of the inner face	0.01873 mm (±3 %)

Heat on the same model: Heat step → new steady-state case: inner face **Temperature** 100 °C, outer face (R 40) 20 °C. Results (case "Heat: …"), S235, λ = 53.3 W/(m·K):

Quantity	Expected value
Temperature at r = 30 mm	53.2 °C (logarithmic profile)
Heat flux |q| at r = 30 mm	≈ 205 000 W/m²
Heat through the inner face	≈ 96.6 W (boundary condition heat table), balance within 0.1 %

T4 – shaft: torsion through a remote point

Shaft Ø20 × 200 mm, fixed at x = 0, torque 100 N·m on the other end.

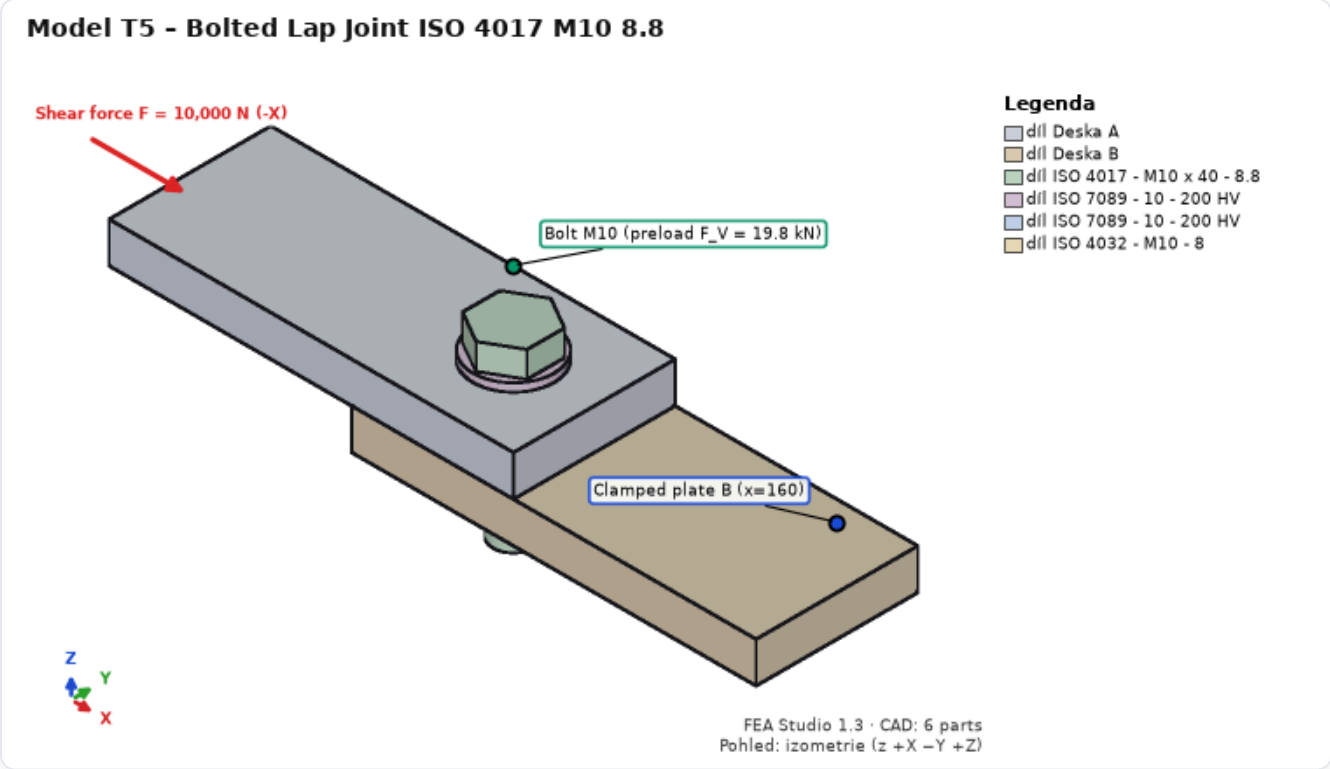


1. Open `T4_shaft.step`, Steel S235; fixed support on face x = 0.
2. Loads: face x = 200 → **Remote force / moment**, point (200; 0; 0), moment Mx = 100,000 N·mm, distributing coupling.
3. Mesh 2.5 mm; run.

Quantity	Expected value
Shear stress at the surface (τ = T/Wk)	63.66 MPa; von Mises ≈ √3 · 63.66 = 110 MPa
Twist of the end	0.903° (end circumference ≈ 0.158 mm)
Reaction moment at the support	100,000 N·mm

T5 – M10 bolted joint: detection, preload, checks

Two lap-jointed 10 mm plates, ISO 4017 M10 8.8 bolt with washers and nut (names from CAD).

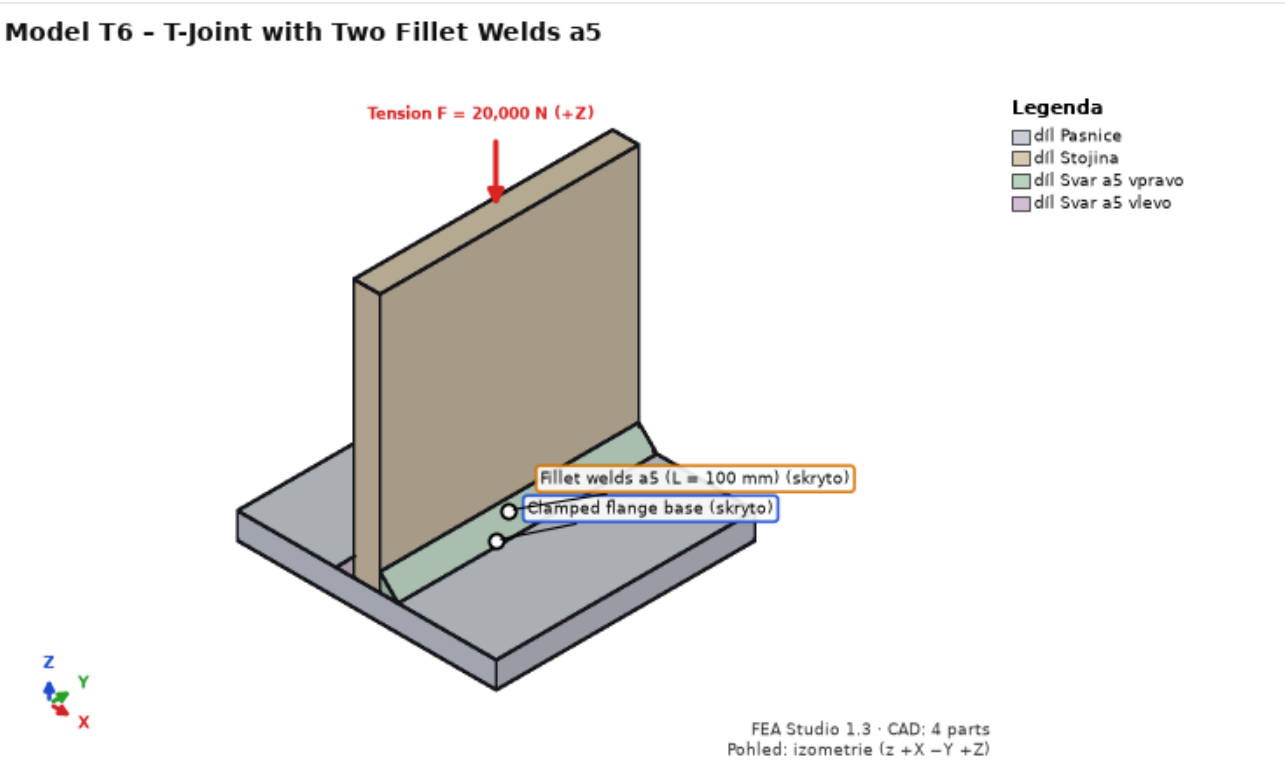


1. Open `T5_joint_M10.step`, plates Steel S235.
2. Joints → **Detect joints…** → confirm the bolt (M10 8.8, with nut, washers).
3. Supports: end face of plate B at x = 160 → Fixed support. Loads: end face of plate A at x = 0 → Force (−10,000; 0; 0) N.
4. Mesh 5 mm; run (CalculiX – other solvers do not support bolts).

Quantity	Expected value	Where
Bolt preload case – bolt force	F_V = 19 760 N	Joints – utilization dock
VDI 2230 F_M,zul (μ_G = 0.12)	29 603 N	bolt check
EN 1993-1-8 F_v,Rd (shear in thread)	22 268 N	bolt check
EN 1993-1-8 F_b,Rd (bearing, e1 = e2 = 20)	43 636 N	bolt check
Shear through interface + bolt	10,000 N	joint forces

T6 – T-joint with a5 fillet welds

Flange 10 mm, web 10 × 100 mm, two a5 fillet welds of length 100 mm (bodies “Svar …”).

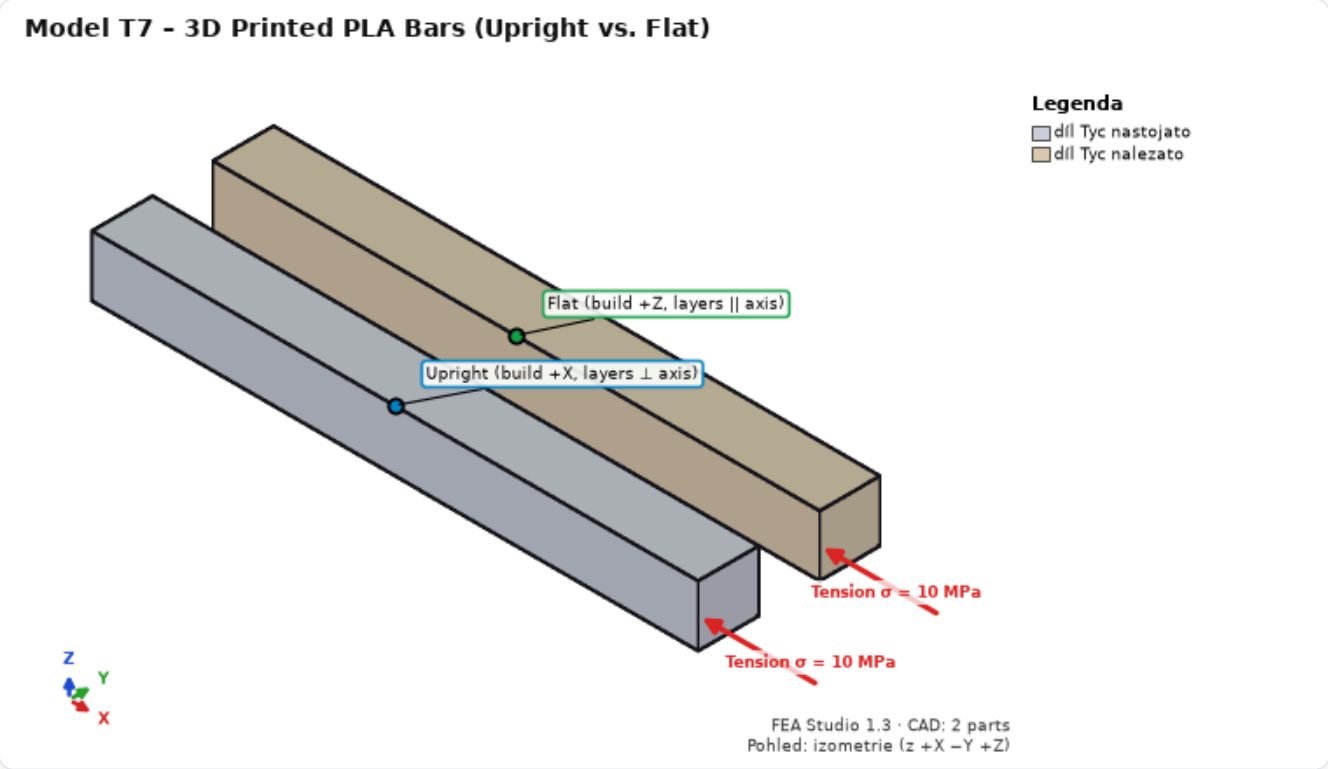


1. Open `T6_t_joint_welds_a5.step`, Steel S235. Joints → Detect joints… → confirm 2 welds.
2. Supports: bottom face of the flange (z = −10) → Fixed support.
3. Two load cases: "Tah" (tension): top face of the web (z = 100) → Force (0; 0; 20,000) N; "Podélný smyk" (longitudinal shear): same face → Remote force at (0; 50; 0), force (0; 10,000; 0) N.
4. Mesh 5 mm; run.

Quantity	Expected value
Effective throat a / weld length from the mesh	5.0 mm / 100 mm
Tension: sum of segment forces of both welds	20,000 N
Longitudinal shear: mean $\tau_{\parallel} = F / (2 \cdot a \cdot L)$	10 MPa

T7 – 3D printing: orthotropy by build direction

Two 100 × 10 × 10 mm PLA bars: one printed upright (layers perpendicular to the length), the other lying flat.



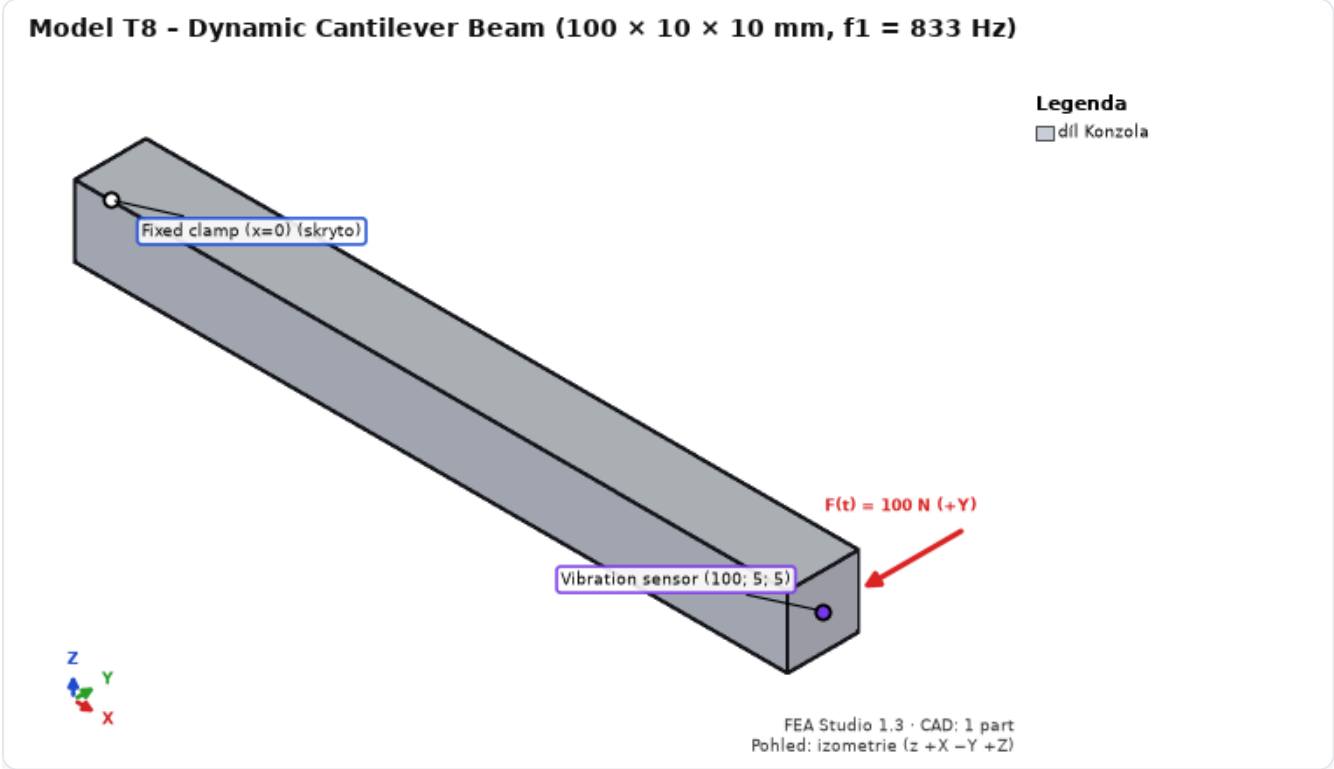
1. Open `T7_printed_bars.step`. Material of both parts **Print PLA Bambu Basic**.
2. 3D printing: part "Tyc nastojato" (upright) – build direction **+X**; "Tyc nalezato" (flat) – **+Z**; for both turn off "shell + infill" (whole part solid) → Apply to selected parts. Failure criterion **Tsai-Hill**.
3. Supports: end faces x = 0 → ux = 0; side faces y = 0 and y = 20 → uy = 0; bottom faces z = 0 → uz = 0.
4. Loads: both end faces x = 100 → Pressure −10 MPa. Mesh 3 mm; run.

Quantity	Expected value
Elongation of the upright bar (perpendicular to layers)	σL/E_z = 0.4630 mm
Elongation of the flat bar (in the layer plane)	σL/E_xy = 0.3731 mm
Safety factor (Tsai-Hill)	upright 3.50 (Rm_z/σ), flat 3.90 (Rm/σ)
Reaction Fx	2,000 N

Then turn on "shell + infill" (3 perimeters, infill 20 %): stiffness and safety both drop; the Shell / core section shows the solid shell share.

T8 – cantilever: natural frequencies and harmonic response

Steel beam 100 × 10 × 10 mm fixed at one end, harmonic force 100 N at the free end (verification Y2).



1. Open `T8_cantilever_10x10x100.step`, Steel S235.
2. Supports: face x = 0 → **Fixed support**. Loads: face x = 100 → **Force** (0; 100; 0) N, load case named "Síla Y 100 N".
3. Step **Dynamics**: **compute dynamics**, 10 modes; modal damping ζ = 2 %; **compute harmonic response**, 10–2 000 Hz; excitation **force** – load case Síla Y 100 N (Add excitation); probe at (100; 5; 5).
4. Mesh 2.5 mm; run (about 20 s).

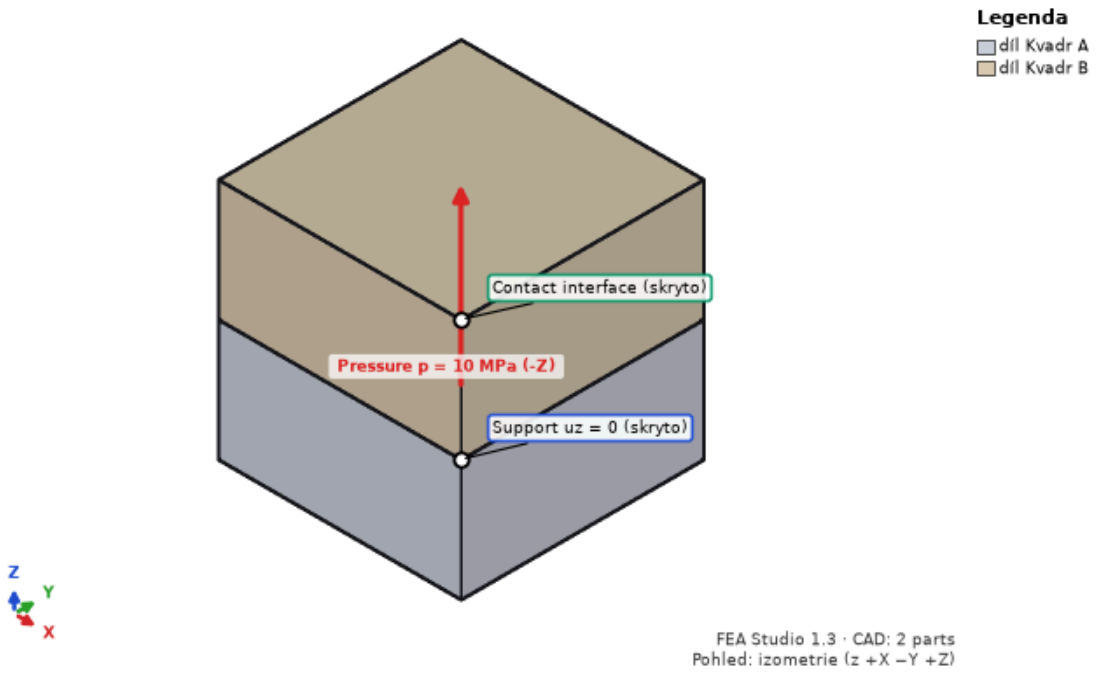
Quantity	Expected value	Where to read
1st and 2nd natural frequency (bending in y and z)	833 Hz (Euler–Bernoulli 835.5 Hz, Timoshenko 0.3 % lower; ±1 %)	Dynamics – table of modes
Sum of effective masses of both modes in y	61 % (E–B 61.3 %); a square section may split it between the two modes	table of modes
3rd and 4th mode (2nd bending), 5th (torsion), 6th (axial)	≈ 5,000 Hz (E–B 5,236 Hz), ≈ 7,390 Hz, ≈ 12,970 Hz	table of modes
Static tip deflection (Síla Y 100 N)	0.190 mm (E–B $FL^3/3EI$ = 0.1905 mm)	Displacement Y, summary
Peak $\|u_y\|$ at the probe (resonance)	4.62 mm = 24.2 × static (E–B 0.9707/(2ζ) = 24.27; ±2 %)	Dynamics – FRF plot, probe
$\|u_y\|$ at 10 Hz	0.190 mm (quasi-static)	FRF plot

With ζ = 1 % the peak doubles (resonance response ∝ 1/damping).

T9 – contact of two blocks: contact pressure

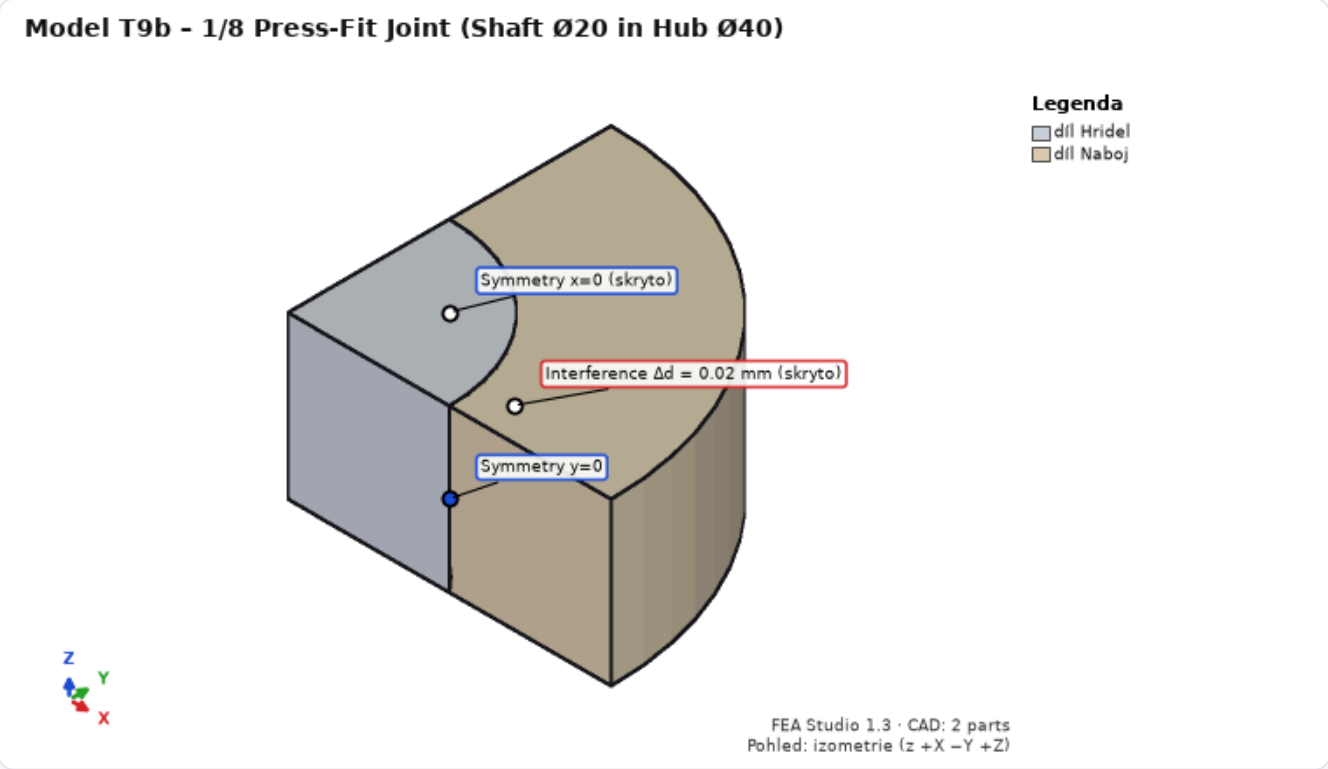
Two steel blocks 20 × 20 × 10 mm stacked, 10 MPa on the top face, frictionless contact (verification K1, "patch test").

Model T9 - Contact Pair Between Two Blocks (20 × 20 × 10 mm)



1. Open `T9_contact_blocks.step`, Steel S235.
2. Supports (sliding): bottom face z = 0 → uz = 0; faces x = 0 (both blocks) → ux = 0; faces y = 0 → uy = 0.
3. Loads: top face z = 20 → **Pressure** 10 MPa.
4. Step **Contacts**: **Find touching parts** → pair Kvadr A – Kvadr B, **Frictionless contact** → Add. The analysis is nonlinear (CalculiX).
5. Mesh 2.5 mm; run (about 20 s).

Quantity	Expected value	Where to read
Contact – pressure	10.0 MPa uniform (±0.5 %)	Contact – pressure
Normal force of the pair	4,000 N (10 MPa × 400 mm^2)	Contacts tab
Displacement of the top face	−0.000953 mm (2ph/E + p/K, K = 50·E)	Displacement Z
Contact – opening, status	0 mm, whole face in contact	contact quantities

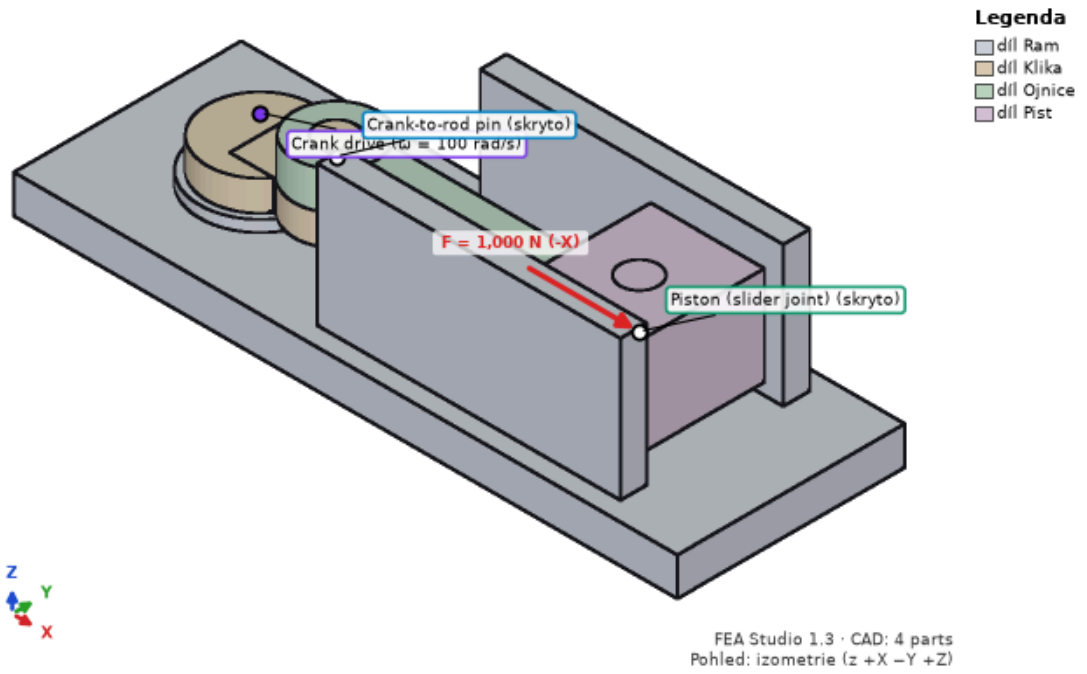


T9b – press fit (`T9b_press_fit_eighth.step` , K4): one eighth of a Ø20 shaft in a Ø20/40 hub, length 2 × 10 mm. Symmetry supports x = 0 → ux = 0, y = 0 → uy = 0, z = 0 → uz = 0. Empty load case “Lis”. Contacts: pair Hridel – Naboj, **Interference (press fit)**, interference −0.01 mm (Δd = 0.02 mm). Mesh 1.25 mm (about 1 min). Expected (Lamé): contact pressure 78.75 MPa (FEA mean ~80 MPa), hub hoop stress at the bore 131.3 MPa, radial displacement of the hub bore +7.4 µm, of the shaft −2.6 µm.

T10 – crank mechanism: kinematics and drive torque

Crank r = 20 mm, connecting rod l = 80 mm, piston; crank at 100 rad/s; 1,000 N on the piston against the X direction (verification MB1).

Model T10 - Slider-Crank Mechanism (Frame, Crank r=20, Rod l=80, Piston)



1. Open `T10_crank_r20_l80.step`, Steel S235 for all parts.
2. Step **Motion**, tab **Bodies**: **Body for each part** (the first part Ram = frame).
3. Tab **Joints**: **Suggest from geometry** → accept 3 revolute joints (frame–crank, crank–rod, rod–piston). Add a **Slider** frame – piston, origin (100; 0; 13), axis (1; 0; 0).
4. Tab **Drivers and forces**: driver of frame–crank, **angular velocity** 100 rad/s; load on body Píst, force (−1,000; 0; 0) N at (100; 0; 13); gravity off.
5. Tab **Analysis**: kinematic, 0.0628319 s (one revolution), 360 outputs → **DOF check** (0 DOF with drivers, 3 redundant constraints) → **Run motion**.

Quantity at t = 0.010472 s (crank at 60°)	Expected value
Piston position x (centre of mass = pin axis)	88.102 mm ($r\cdot\cos\varphi + \sqrt{(l^2 - r^2\cdot\sin^2\varphi)}$)
Piston velocity v_x	−1,953.8 mm/s
Piston acceleration a_x	−75,022 mm/s²
Drive torque without inertia (set mass 0.001 kg and zero inertia manually)	$\|M\| = F\cdot\|v\|/\omega$ = 19,538 N·mm
Drive torque with CAD mass (steel)	$\|M\| \approx$ 19,250 N·mm (inertia of rod and piston)
Body equilibrium, power balance	≈ 0 (motion result summary)

The sign of velocity and torque depends on the crank joint axis (+z = counter-clockwise seen from above). Then try **Transfer to FEA...** for the connecting rod (instant of maximum pin reaction), as in example 8.

What else to try

- **Solvers:** Solve → Solver section: CalculiX / NGSolve + PARDISO / NGSolve + hypre; T1 displacement must agree to 4 significant digits. NGSolve solvers come with the **Full** profile (modules ngsolve, hypre, and on Windows

Microsoft MPI); in the standard install they are greyed out (“not installed”) – install with **Install solvers…** (Tools → Program modules…) and restart, or skip.

- **Undo / Redo** (`Ctrl+Z` / `Ctrl+Y`); save `Ctrl+S` and reopen (results are saved with the project).
- **Word report** (`Ctrl+P`) – compare the numbers with the tables.
- **Section** in results (top bar: Section → axis and slider) and a **probe** by clicking into the model.
- **Automation:** command line, MCP, REST, e.g. `python -m fea_core run study.json`.

Reporting bugs

Send the saved `.fea` project (Ctrl+S), the text of the **Solver log** dock (Ctrl+A, Ctrl+C), what you expected and what happened; for a crash also the command-line window contents, if open.